QWeatherRiskF

Issue 1: Crop data validation report

Item	Detail
Input (unchanged)	merged_cleaned_v1d.csv, 65 rows x 151 columns (13 districts x 2015-2019)
Output	crop_long_v1.csv, 1,430 rows x 120 columns (65 district-years x 22 crops)
Script	crop_restructure_validate.py (reproduces everything in this report)
Date	November 2026

1. Summary

- The wide dataset was restructured into crop-long format: **one row per district-year-crop, 22 crops, 1,430 rows, no duplicate keys**. All 4,290 source crop values (66 columns x 65 rows) were compared cell by cell with the long file and are **identical**; all 78 weather columns and the identifiers are preserved unchanged on every row; the input file's checksum is unchanged.
- **No missing values remain in the crop data** (v1d filled them), but 126 crop-district-years (365 cells) were blanks in the earlier version that were converted to 0. They are flagged (flag_nan_filled_with_zero) so the origin is not lost.
- **Two aggregate crops and two seasonal components are retained and flagged.** Sorghum equals the sum of kharif and rabi sorghum (all 65 district-years). Oilseeds does **not** equal the sum of its eight member crops in 35 of 65 district-years (seven districts, every year).
- Data-quality flags identify 379 zero-area rows, 36 rows with area but zero production, 6 with production but zero area, 89 rows where reported yield differs by more than 5% from production/area, and 70 sharp year-to-year yield jumps. Known problems (cotton 2019, minor pulses 2018-19, sugarcane correction) are flagged and described below.

2. What changed between v1c and the v1d input

Because the long file is built from v1d, I compared v1d against my corrected v1c file. The weather columns, identifiers and all non-blank crop values are identical. The differences are:

- Column names converted to snake_case (e.g. RICE YIELD (Kg per ha) became rice_yield_kg_per_ha).
- Barley (3 columns) and the six area-only series (fruits, vegetables, fruits and vegetables, potatoes, onion, fodder) were dropped, leaving 22 crops with complete area, production and yield.
- All 365 blank crop cells were filled with 0, **including yield**. For rows where area is 0, yield is mathematically undefined, so a stored 0 yield is not a true observation. The rows are flagged (area_zero, yield_zero) so later steps can treat yield as undefined there.
- Not changed in v1d: minor pulses 2018-19 (still 0), cotton 2019 (still the suspect low values), the oilseeds aggregate, the sorghum aggregate, and all 78 weather columns including rabi_prev, annual_cal and kharif cold_days. The sugarcane correction from v1c is retained.

3. Method

Column identification. Crop columns were found with the pattern crop_{area|production|yield}_{unit}. 22 crops have all three measures (66 columns, none partial). The other 85 columns are 6 identifier/metadata columns plus rain_valid_area_fraction (7) and 78 weather columns (names containing a double underscore). Full list: validation_tables/column_inventory.csv.

Restructuring. Each crop was stacked into rows keyed by district_code, year and crop; area, production and yield were copied without change; the district-year identifiers and all weather features were attached to every crop row (many-to-one merge, validated). Weather is therefore repeated across the 22 crop rows of a district-year, which is intended for crop-level modelling.

Output columns (120). Keys and labels (district_code, district_name, year, crop, crop_name, crop_group); role indicators (is_aggregate, is_seasonal_component, season_component, component_of_aggregate,

is_group_category); original measures (area_1000_ha, production_1000_tons, yield_kg_per_ha); derived checks (yield_check_kg_per_ha, yield_rel_diff, yield_vs_other_years_median, aggregate_member_sum_area/production); 17 flag columns plus n_flags and flag_summary; then state/crop-district metadata, rain_valid_area_fraction and the 78 weather columns.

4. Acceptance checks

Type	Check	Result	Detail
structural	Every crop has area, production and yield columns	PASS	22 crops x 3 = 66 columns; partial: []
structural	All columns classified (crop / weather / identifier-metadata)	PASS	66 crop + 78 weather + 7 id/meta = 151
structural	Source key unique (district_code, year)	PASS	65 rows
structural	One row per district-year-crop	PASS	1430 rows = 65 district-years x 22 crops
structural	No duplicate keys (district_code, year, crop)	PASS	0 duplicates
structural	Round trip: long values identical to every source crop cell	PASS	4290 cells compared exactly
structural	Weather features preserved unchanged (all rows)	PASS	78 weather columns
structural	Weather identical across crops within a district-year	PASS	
structural	Source file unchanged (SHA-256 before == after)	PASS	d888554eac5f269a...
data-quality finding	Negative values	OK	0 rows
data-quality finding	Missing values in crop columns	OK	0 NaN cells (v1d has none)
data-quality finding	Aggregate "sorghum" equals sum of its members	OK	0 of 65 district-years mismatch (tolerance 5% or 0.1)
data-quality finding	Aggregate "oilseeds" equals sum of its members	FLAGGED	35 of 65 district-years mismatch (tolerance 5% or 0.1)
data-quality finding	Yield reported = production/area x 1000 (area>0, prod>0)	FLAGGED	89 rows differ by more than 5%
data-quality finding	Zero area with positive production	FLAGGED	6 rows (rounding of tiny areas)
data-quality finding	Positive area with zero production	FLAGGED	36 rows

Structural checks must pass for the script to finish. Data-quality findings are results to document and do not stop the script.

Acceptance criteria from the issue: one record per district-year-crop (PASS); no duplicate keys (PASS); original values and source unchanged (PASS: exact round trip and unchanged checksum); aggregate crops and missing observations documented (see sections 5 and 6).

5. Aggregate crops, seasonal components and group categories

Crop	Role in long file	Check	Result
sorghum	Aggregate (is_aggregate = True)	Equals kharif_sorghum + rabi_sorghum for area and production	Consistent in 65 of 65 district-years
kharif_sorghum, rabi_sorghum	Seasonal components (is_seasonal_component = True; season_component = kharif / rabi; component_of_aggregate = sorghum)	Sum equals aggregate	Consistent
oilseeds	Aggregate (is_aggregate = True)	Compared with the sum of groundnut, sesamum, rapeseed and mustard, safflower, castor, linseed, sunflower and soyabean	35 of 65 district-years differ by more than 5% or 0.1 (flag_aggregate_ mismatch); see Figure 1
the 8 oilseed crops	component_of_aggregate = oilseeds	Retained as separate crops	Retained
minor_pulses	Group category (is_group_category = True)	Members are not in the dataset, so no sum check is possible	Not verifiable

Important for modelling: an aggregate and its components must not be used together as separate crops, because that double counts the same land and production. The flags allow either set to be selected or excluded.

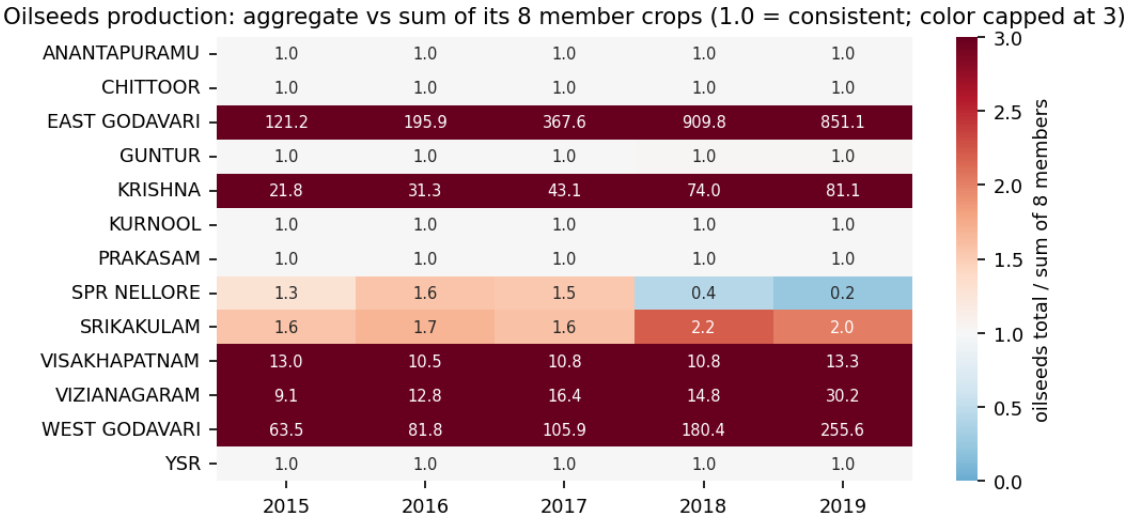


Figure 1. The oilseeds total is many times the sum of its eight members in East and West Godavari, Krishna, Visakhapatnam and Vizianagaram (about 1.6-2.2 times in Srikakulam), and is below the sum in Nellore in 2018-19. The seven affected districts are flagged in all five years. The cause is unconfirmed (the total probably includes crops not in the dataset, such as oil palm or coconut).

6. Missing, zero and inconsistent observations

Missing. No NaN values remain in crop columns. 126 rows (365 cells) were blank in v1c and are 0 in v1d; they are mostly wheat, linseed, safflower, soyabean, rapeseed and mustard, castor and finger millet (Figure 2).

Zero. 379 rows have zero area; 359 have zero area, production and yield. Zero yield with positive area occurs in 8 rows (linseed 5, chickpea, kharif sorghum and rapeseed 1 each).

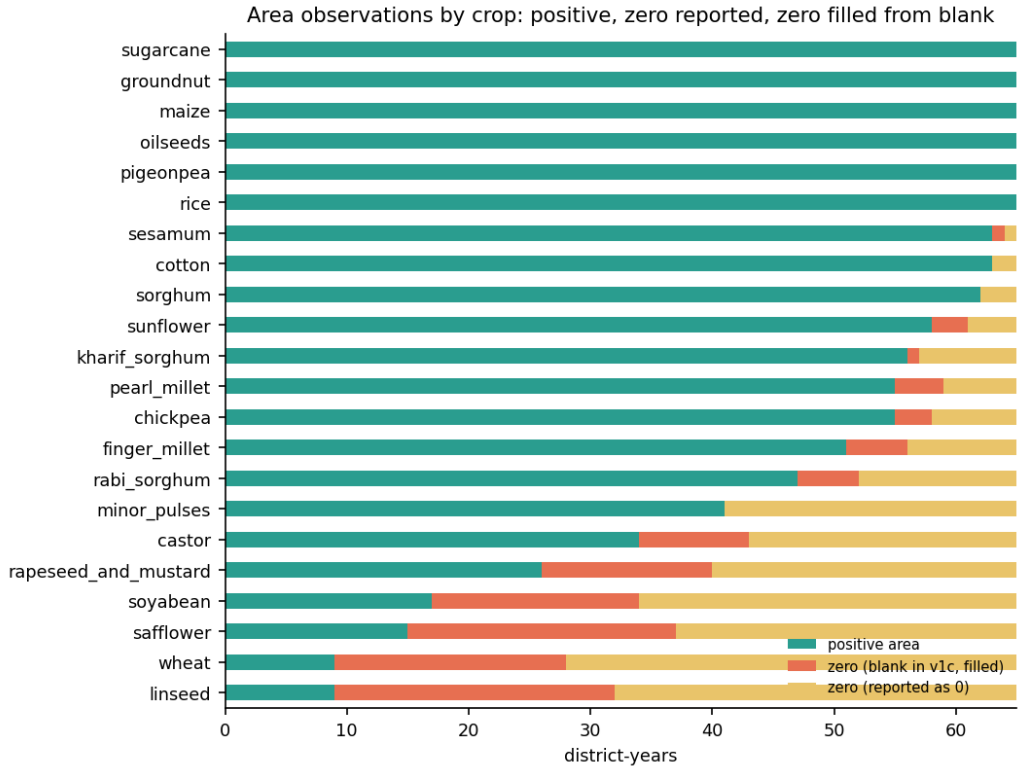


Figure 2. Area observations by crop. Wheat, linseed, safflower, soyabean and rapeseed/mustard have positive area in 9-26 of 65 rows; the red portion is zero values that were blank in the earlier version.

Inconsistent. Reported yield differs from production / area x 1000 by more than 5% in 89 rows, mostly cotton (20) and tiny-area crops where production is rounded to whole or tenth thousand tonnes. 36 rows have area but zero production (cotton 8, rapeseed 8, linseed 6, others) and 6 have production with zero area, both consistent with rounding of very small values. 70 rows have yield below half or above double the same district-crop median in other

years (cotton 13, oilseeds 11, chickpea 8, rabi sorghum 7, pigeonpea 7), largely the 2018 drought and the cotton 2019 drop.

Known issues, flagged per row

- **flag_sugarcane_unit_corrected** (39 rows, sugarcane 2015-17): production and yield were multiplied by 10 in v1c after verification against official totals. No sugarcane yield jump remains (0 flagged jumps).
- **flag_cotton_2019_suspect** (13 rows): cotton production and yield in 2019 fall to about a fifth of earlier years in most districts with area unchanged. Not corrected; cause unconfirmed.
- **flag_minor_pulses_not_reported** (24 rows): minor pulses area and production are 0 in 12 districts in 2018-19 after 600-740 thousand ha in 2015-17. These zeros mean "not reported", not "none grown".
- **flag_low_area_lt_1000ha** (351 rows): area below 1,000 ha, where yield is unreliable because of rounding.

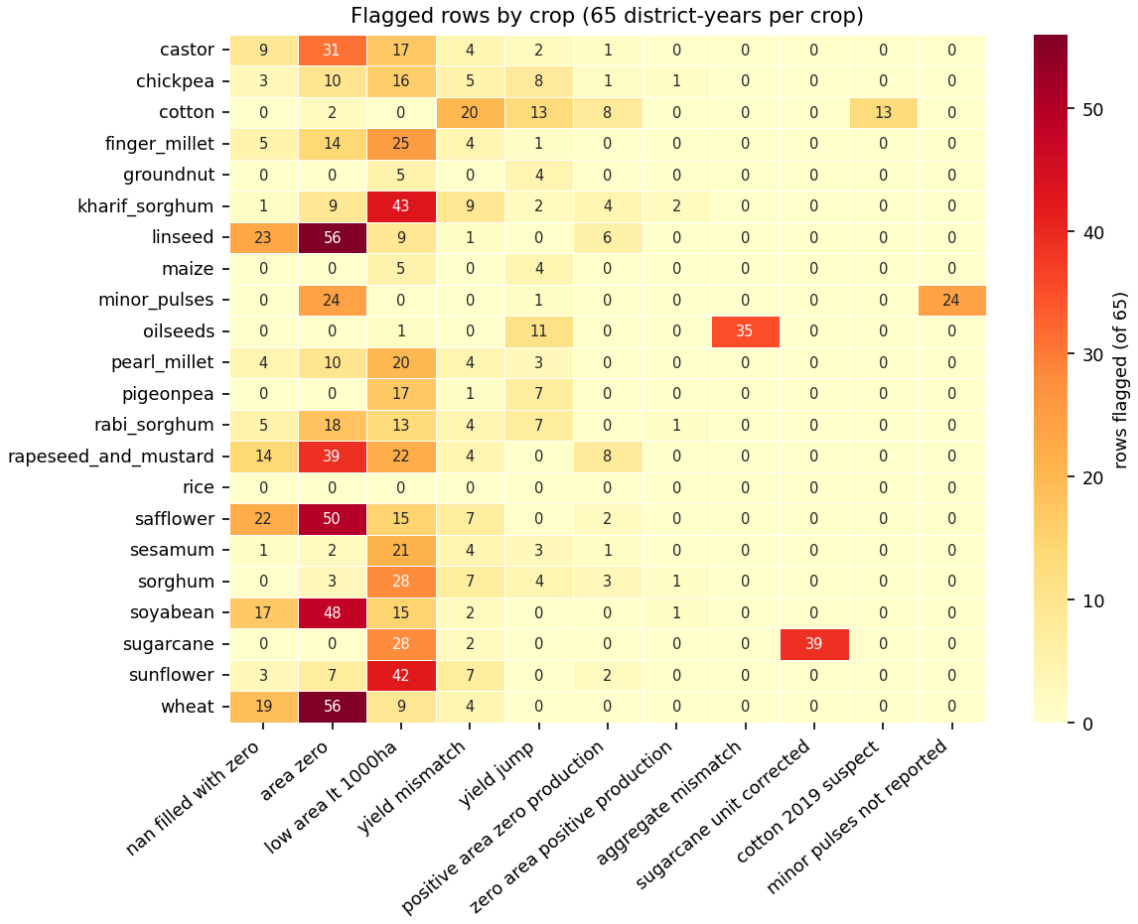


Figure 3. Number of flagged rows per crop (out of 65). Rice has no flags; the small oilseed and cereal crops and cotton carry the most.

7. Limitations and open points

- Yield is 0, not undefined, where area is 0 (set in v1d). The flags mark these rows; the long file keeps the v1d values as required, and the next step should decide whether to treat them as missing for modelling.
- The oilseeds, cotton 2019 and minor pulses problems are flagged but not resolved; they need official figures or the data owner.
- crop_group is a descriptive grouping I assigned for convenience and is not from the source data.
- The long file repeats the 78 weather columns for every crop row, so it is wide (120 columns). A slimmer variant with only keys and crop columns can be derived by dropping those columns.
- Weather windows and units are still undocumented, as noted in earlier reports.

8. How to reproduce

python crop_restructure_validate.py --input merged_cleaned_v1d.csv --reference merged_district_year_v1c.csv --outdir .
The --reference file is optional; without it flag_nan_filled_with_zero is all False. Outputs: crop_long_v1.csv and validation_tables/*.csv.